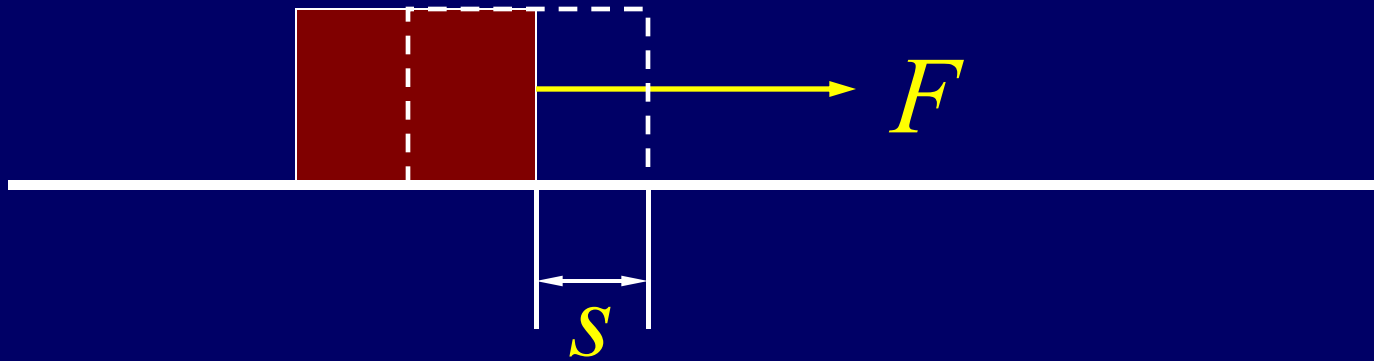


Chapter 11

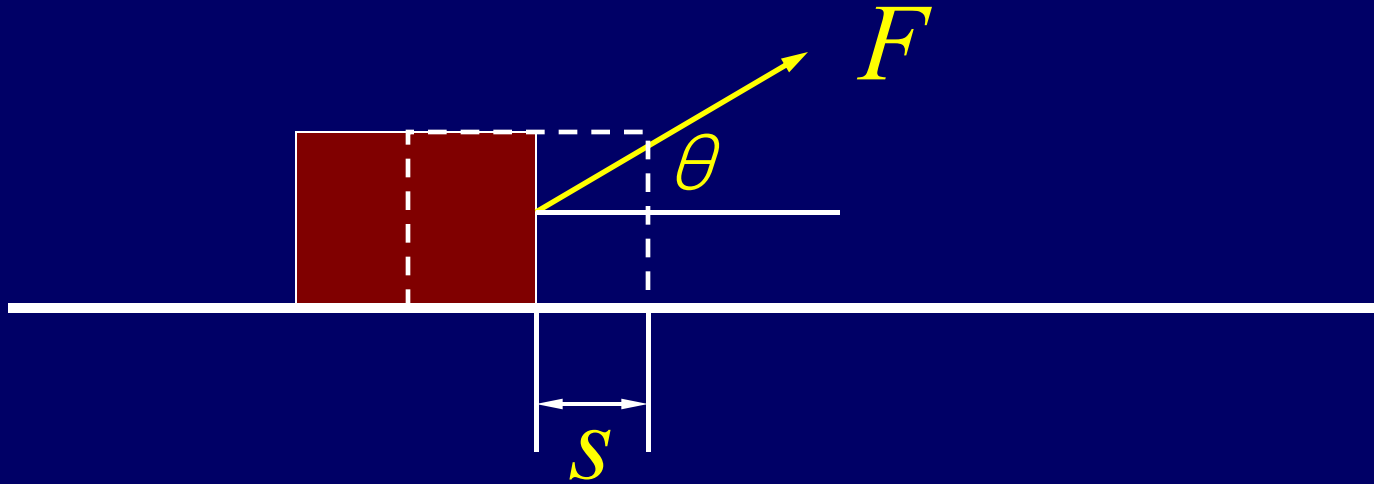
Work and Kinetic Energy

Work done by a constant force



$$W = Fs$$

Work done by a constant force



$$W = Fs \cos \theta$$

$$W = \vec{F} \cdot \vec{s} \quad dW = \vec{F} \cdot d\vec{s}$$

Power

$$\bar{P} = \frac{W}{\Delta t}$$

$$P = \frac{dW}{dt}$$

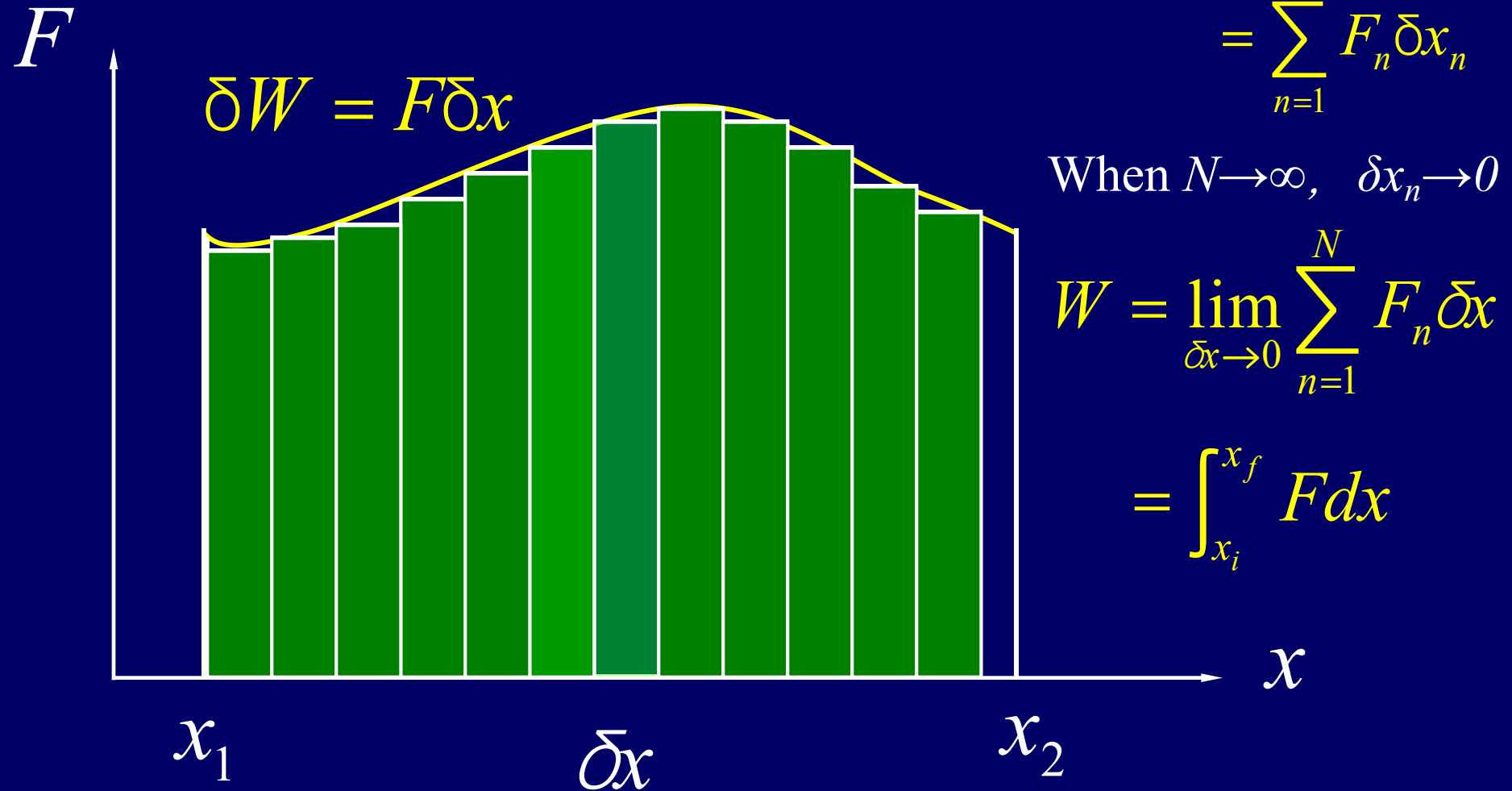
$$P = \frac{\vec{F} \cdot d\vec{s}}{dt} = \vec{F} \cdot \frac{d\vec{s}}{dt} = \vec{F} \cdot \vec{v}$$

The unit of power in SI unit is the joule per second, which is called the watt (W).

Work done by a variable force .(One dimensional case)

$$W = \delta w_1 + \delta w_2 + \delta w_3 + \cdots = F_1 \delta x_1 + F_2 \delta x_2 + F_3 \delta x_3 + \cdots$$

$$= \sum_{n=1}^N F_n \delta x_n$$

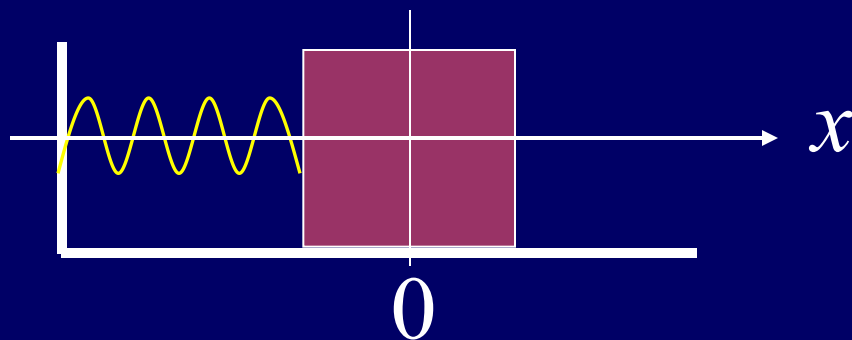


$$W = \lim_{\delta x \rightarrow 0} \sum_{n=1}^N F_n \delta x$$

$$= \int_{x_i}^{x_f} F dx$$

Example:

$$F = -kx.$$



$$W = \int_{x_i}^{x_f} F dx = \int_{x_i}^{x_f} -kx dx = \frac{1}{2} kx_f^2 - \frac{1}{2} kx_i^2$$

Two dimensional case

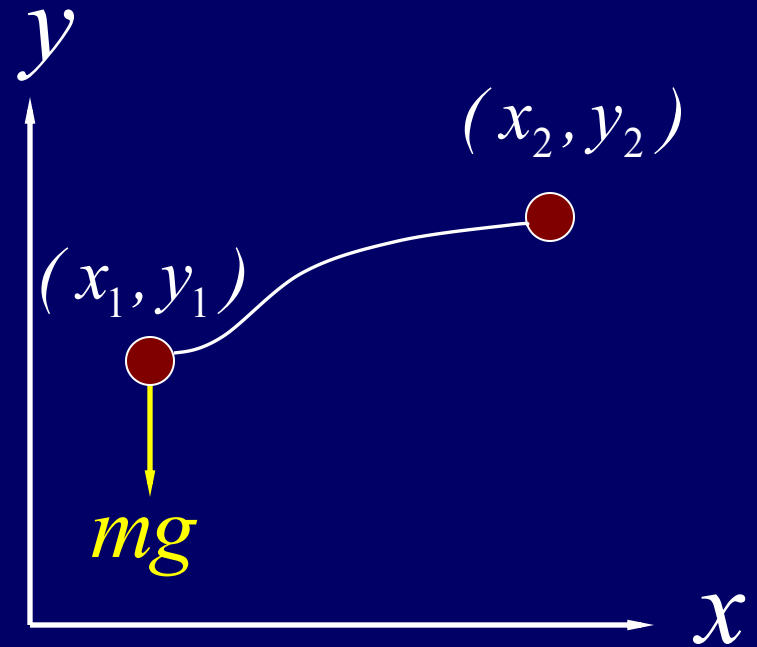
$$W = \int_i^f \vec{F} \cdot d\vec{s}$$

$$W = \int_i^f (F_x \vec{i} + F_y \vec{j}) \cdot (dx \vec{i} + dy \vec{j})$$

$$= \int_i^f F_x dx + F_y dy$$

$$= \int_{y_1}^{y_2} -mg dy$$

$$= mg(y_1 - y_2)$$



Kinetic energy and work-energy theorem

$$dW = \vec{F} \cdot d\vec{s}$$

$$dW = m\vec{a} \cdot d\vec{s} = m \frac{d\vec{v}}{dt} \cdot d\vec{s} = m\vec{v} \cdot d\vec{v}$$

$$W = \int dW = \int m\vec{v} \cdot d\vec{v}$$

$$= \frac{1}{2}mv^2 \Big|_{v_1}^{v_2} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \Delta K$$

Kinetic energy

$$\int_0^t \vec{F} dt = \Delta \vec{p}$$

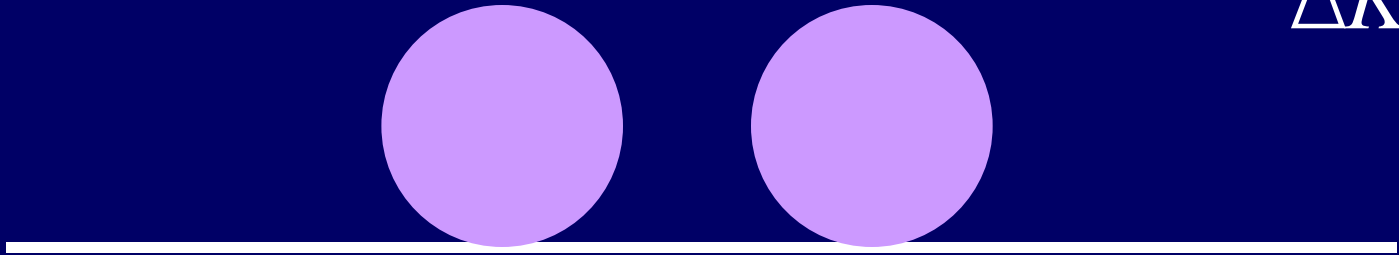
Kinetic energy in collisions

if $\Sigma \vec{F}_{ext} = 0$, $\vec{p} = \text{const.}$

Elastic collisions

if $\Sigma \vec{F}_{ext} = 0$, $\Delta K = ?$

$$\Delta K = 0$$



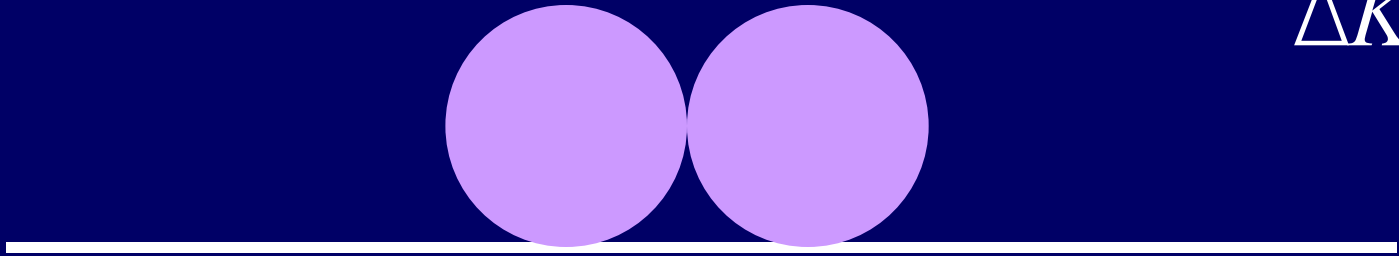
Kinetic energy in collisions

$$\text{if } \Sigma \vec{F}_{ext} = 0, \vec{p} = \text{const.}$$

Inelastic collisions

$$\text{if } \Sigma \vec{F}_{ext} = 0, \Delta K = ?$$

$$\Delta K < 0$$



$$\Sigma \vec{F}_{int} \cdot d\vec{s} \neq 0, W \neq 0$$

$$K_i \geq K_f$$

$$\Delta K \neq 0$$

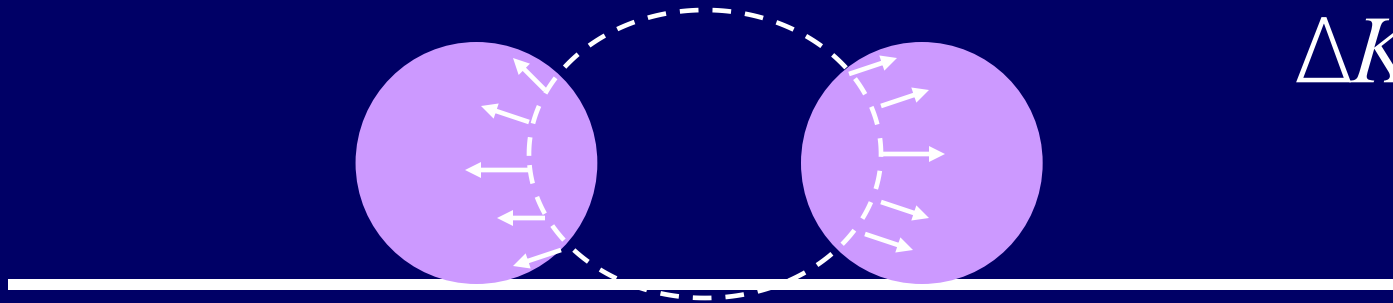
Kinetic energy in collisions

if $\Sigma \vec{F}_{ext} = 0$, $\vec{p} = \text{const.}$

Explosive

if $\Sigma \vec{F}_{ext} = 0$, $\Delta K = ?$

$\Delta K > 0$

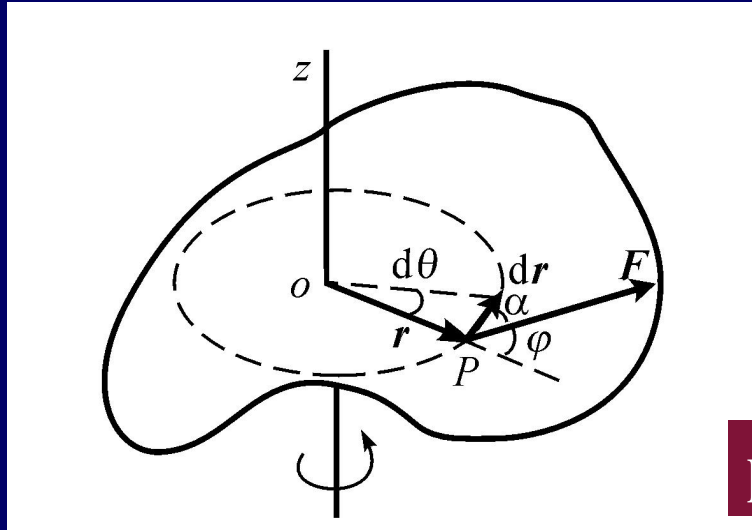


$\Sigma \vec{F}_{int} \cdot d\vec{s} \neq 0$, $W \neq 0$

$$K_i \leq K_f$$

$\Delta K \neq 0$

Work and kinetic energy in **rotational motion**



power

Rotational kinetic energy

$$dW = \vec{F} \cdot d\vec{s}$$

$$= F \sin \phi (r d\theta)$$

$$= (r F \sin \phi) d\theta = \tau_z d\theta$$

$$W = \int_{\theta_i}^{\theta_f} \tau_z d\theta$$

work

$$P = \frac{dW}{dt} = \frac{\tau_z d\theta}{dt} = \tau_z \omega_z$$

$$K = \sum \frac{1}{2} m_n v_n^2 = \sum \frac{1}{2} m_n (r_n \omega)^2 = \frac{1}{2} \sum m_n r_n^2 \omega^2 = \frac{1}{2} I \omega^2$$

Review

Translational Quantity

Rotational Quantity

Work

$$W = \int \vec{F} \cdot d\vec{s}$$

$$W = \int \tau \cdot d\theta$$

Power

$$P = \vec{F} \cdot \vec{v}$$

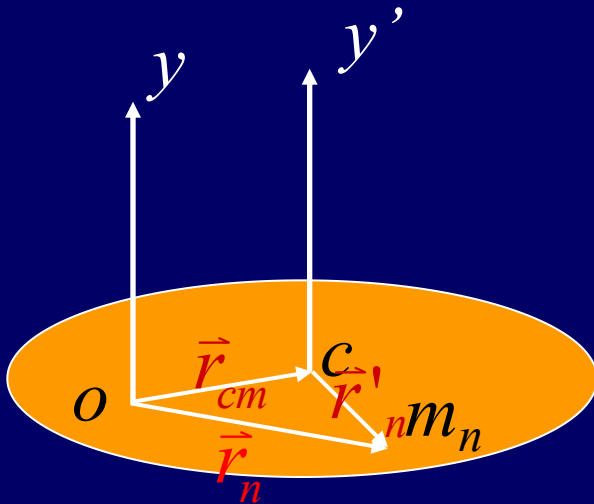
$$P = \vec{\tau} \cdot \vec{\omega}$$

Kinetic
energy

$$K = \frac{1}{2}mv^2$$

$$K = \frac{1}{2}I\omega^2$$

Translation and rotation



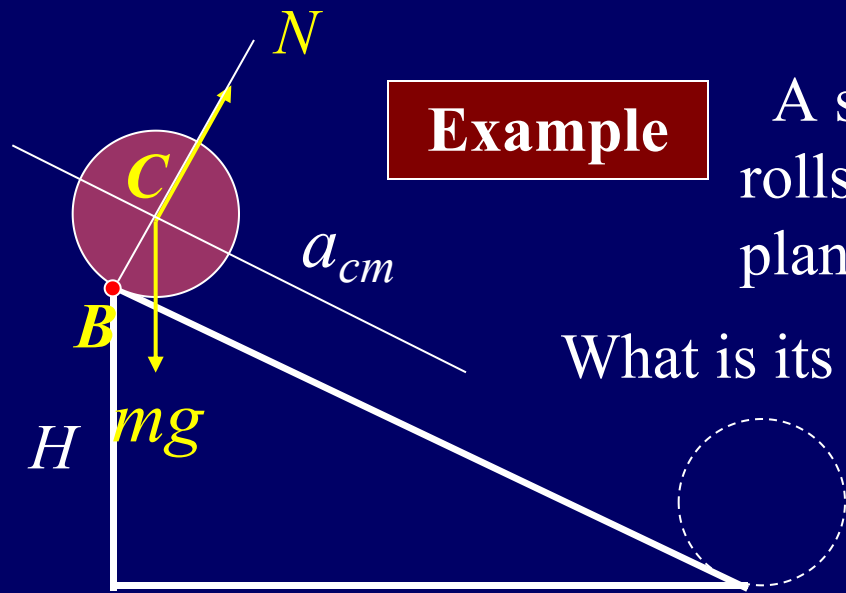
$$K = \sum \frac{1}{2} m_n v_n^2$$

$$K = \sum \frac{1}{2} m_n (\vec{v}_{cm} + \vec{v}'_n)^2$$

$$= \frac{1}{2} \sum m_n v_{cm}^2 + \frac{1}{2} \sum m_n v_n'^2 + \sum m_n \vec{v}_{cm} \cdot \vec{v}'_n$$

$$= \frac{1}{2} M v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2 + 0 \dots$$

$$\frac{1}{2} I \omega^2 = k = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$



Example

A solid cylinder of mass m and radius R rolls **without slipping** down an inclined plane with height H .

What is its kinetic energy?

$$v_{cm}^2 = 2a_{cm}L = \frac{4g}{3}H$$

$$K_T = \frac{1}{2}mv_{cm}^2 = \frac{2mgH}{3}$$

$$K_R = \frac{1}{2}I_{cm}\omega^2 = \frac{mgH}{3}$$

$$\omega^2 = \frac{v_{cm}^2}{R^2} = \frac{4g}{3R^2}H$$

$$K = K_T + K_R = mgH$$

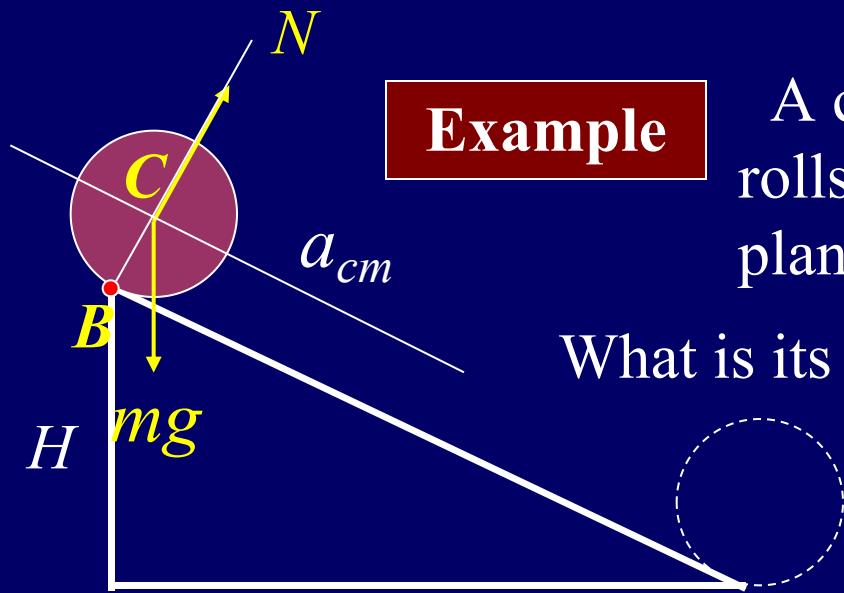
$$K_T : K_R = 2 : 1$$

$$\tau = I\alpha$$

$$Rmg\sin\theta = \left(\frac{3}{2}mR^2\right)\alpha$$

$$\alpha = \frac{2g}{3R}\sin\theta$$

$$a_{cm} = \alpha R = \frac{2g}{3}\sin\theta$$



Example

A cylinder shell of mass m and radius R rolls **without slipping** down an inclined plane with height H .

What is its kinetic energy?

$$v_{cm}^2 = 2a_{cm}L = gH$$

$$K_T = \frac{1}{2}mv_{cm}^2 = \frac{mgH}{2}$$

$$K_R = \frac{1}{2}I_{cm}\omega^2 = \frac{mgH}{2}$$

$$\omega^2 = \frac{v_{cm}^2}{R^2} = \frac{g}{R^2}H$$

$$K = K_T + K_R = mgH$$

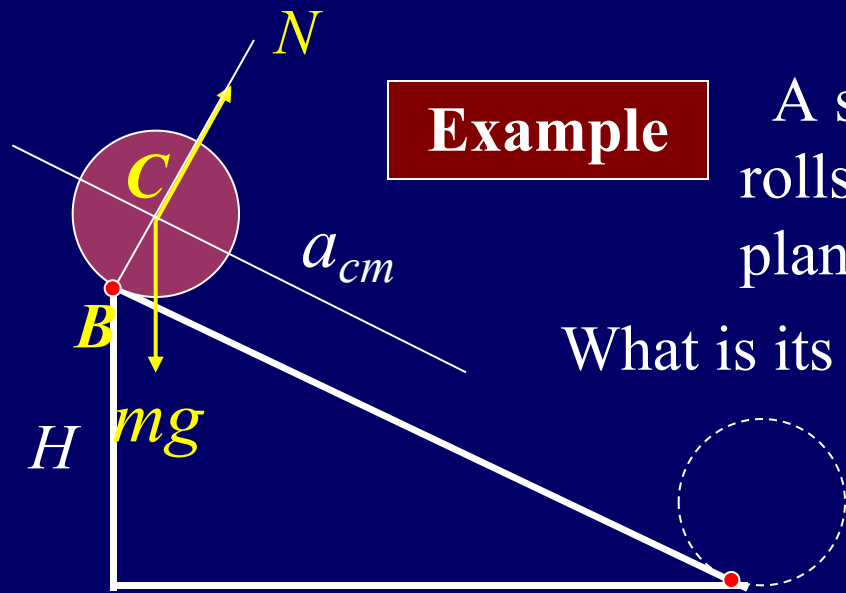
$$K_T : K_R = 1:1$$

$$\tau = I\alpha$$

$$Rmg \sin \theta = (2mR^2)\alpha$$

$$\alpha = \frac{g}{2R} \sin \theta$$

$$a_{cm} = \alpha R = \frac{g}{2} \sin \theta$$



Example

A solid sphere of mass m and radius R rolls **without slipping** down an inclined plane with height H .

What is its kinetic energy?

$$v_{cm}^2 = 2a_{cm}L = \frac{10g}{7}H$$

$$K_T = \frac{1}{2}mv_{cm}^2 = \frac{5mgH}{7}$$

$$K_R = \frac{1}{2}I_{cm}\omega^2 = \frac{2mgH}{7}$$

$$a = \frac{5g}{7R} \sin \theta$$

$$K_T : K_R = 5 : 2 \quad \omega^2 = \frac{v_{cm}^2}{R^2} = \frac{10g}{7R^2}H$$

$$a_{cm} = aR = \frac{5g}{7} \sin \theta$$

$$K_R = \frac{1}{2}I\omega^2 = mgH$$

CHAP. 11

Exercises

P250: 21, 22, 23, 30, 36

Problem

P255: 31, 32